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Research Interests

I am a Ph.D. student in Computer Science at the University of Colorado Boulder. My main research area is **Human-Computer Interaction**. My research goal is to **expand people's capacity to create and learn** by building interactive systems with generative AI and extended reality that adapt content to their actions, context, and intentions. My work spans AI-assisted filmmaking, technology-enhanced learning, and embodied fitness training.

Keywords: human-AI interaction, creativity support tools, adaptive media, embodied interaction

Education

Summary: I trained in fine art, then industrial design, and now study computer science.

August 2024 - **University of Colorado Boulder**
Present Ph.D. student, Computer Science
Supervisor: Ryo Suzuki

September 2020 - **Korea Advanced Institute of Science and Technology (KAIST)**
August 2022 Master of Science, Industrial Design, *Best Master's Thesis Award*
Thesis: Exploring Different Augmented Reality Visualization Methods of Fitness Videos and Their Effects on the Exercise Experience
Supervisor: Andrea Bianchi

March 2011 - **Seoul National University**
February 2016 Bachelor of Fine Arts, Painting and Media Arts, *Cum Laude*
Painting graduation exhibition on fluid identity [A1]
Media Arts graduation exhibition on Korea's [Sampo generation](#) [A4]
Supervisor: Inhwon Oh, Cheol-Woong Sim

Research Experience

Summary: I have built interactive systems in five research groups across academia and industry, including research internships at Autodesk and Fujitsu.

May 2026 - [Autodesk, HCI and Visualization Team](#)
August 2026 Research Intern, Mentors: Frederik Brudy, David Ledo
Created a design space for film directing, grounded in a cinematic corpus, to establish a shared language between human creators and generative AI.

May 2025 - [Fujitsu Research of America, Converging Lab](#)
October 2025 Research Intern, Mentors: Mose Sakashita, Aditi Mishra, Aakar Gupta, Koichiro Niinuma
Built a system that grounds AI video generation in street-view imagery for spatially consistent shots, evaluated with 12 filmmakers [C8].

August 2024 - [University of Colorado Boulder, Programmable Reality Lab](#)

Present

Graduate Research Assistant, Mentor: Ryo Suzuki

Study how people create and consume video by building interactive systems with generative AI, extended reality, and virtual agents for storytelling and education [C7, C8, U1].

January 2023 - [Chung-Ang University, Human-AI Interaction Design & Fabrication Lab](#)

July 2024

Research Assistant, Mentor: Yoonji Kim

Built a filmmaking previsualization tool using image segmentation and deepfake face swapping [C4], and haptic systems that recreate a personal trainer's touch through vibrotactile and electrical stimulation [P1, C6].

September 2020 - [KAIST, Make Lab](#)

September 2022

Graduate Research Assistant, Mentor: Andrea Bianchi

Designed and evaluated augmented reality (AR), virtual reality (VR), and haptic systems for home fitness, remote education, and immersive VR [C1-C3].

Work Experience

Summary: Before research, I spent nearly four years at Dexter Studios, first as a film VFX compositor and later as a VR/AR generalist.

August 2018 - [VR/AR Generalist at Dexter Studios](#)

April 2020

New Media Division, Immersive Content and Virtual Production

Created 3D graphics for VR films, AR exhibitions and apps, and virtual production, covering character design, environments, visual scripting, lighting, and look development [X1-X2, X4].

September 2016 - [Film VFX Compositor at Dexter Studios](#)

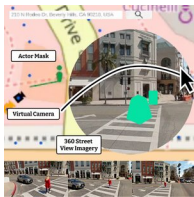
July 2018

Film Post-Production VFX Compositing Team

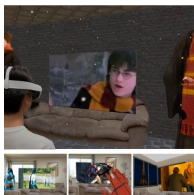
Composited live-action plates, 2D elements, and 3D CG renders into seamless final shots for feature films [F1-F6].

Peer-Reviewed Full Papers

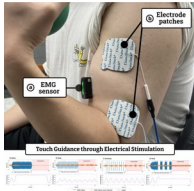
Summary: I have published 8 full papers in CHI (4), UIST (2), ISMAR (1, in IEEE TVCG), and HCIK (1), and have two more under review.



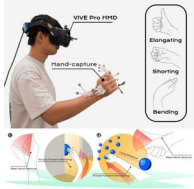
[C8] **Hye-Young Jo**, Mose Sakashita, Aditi Mishra, Ryo Suzuki, Koichiro Niinuma, Aakar Gupta. **Map2Video: Guiding Real-World-Grounded AI Video Generation**. In Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2026. (UIST '26, acceptance rate: 20.1%) Also accepted to the UIST '26 demo track. ([video](#)) ([PDF](#))



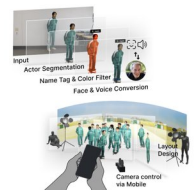
[C7] Keiichi Ihara, DaeHo Lee, Manato Abe, **Hye-Young Jo**, Ryo Suzuki. **CinemaWorld: Generative Augmented Reality with LLMs and 3D Scene Generation for Movie Augmentation**. In Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2026. (UIST '26, acceptance rate: 20.1%) Also accepted to the UIST '26 demo track. ([PDF](#))



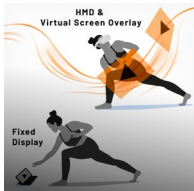
[C6] Dong-Uk Kim, **Hye-Young Jo**, Hankyung Kim, Ryo Suzuki, Seungwoo Je, Yoonji Kim. **TingleTouch: Touch Guidance through Electrical Stimulation in Resistance Training**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2026. (CHI '26, acceptance rate: 25.3%) ([video](#)) ([PDF](#))



[C5] Yilong Lin, Han Shi, Weitao Jiang, Xuesong Zhang, **Hye-Young Jo**, Yoonji Kim, Seungwoo Je. **Designing Hand and Forearm Gestures to Control Virtual Forearm for User-Initiated Forearm Deformation**. IEEE Transactions on Visualization and Computer Graphics. IEEE, 2025. (ISMAR '25, acceptance rate: 7.86%) ([video](#)) ([PDF](#))



[C4] **Hye-Young Jo**, Ryo Suzuki, Yoonji Kim. **CollageVis: Rapid Previsualization Tool for Indie Filmmaking using Video Collages**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2024. (CHI '24, acceptance rate: 26.4%) ([video](#)) ([PDF](#))



[C3] **Hye-Young Jo**, Laurenz Seidel, Michel Pahud, Mike Sinclair, Andrea Bianchi. **FlowAR: How Different Augmented Reality Visualizations of Online Fitness Videos Support Flow for At-Home Yoga Exercises**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2023. (CHI '23, acceptance rate: 27.6%) ([video](#)) ([PDF](#))



[C2] **Hye-Young Jo**, Wooje Chang, Hoonjin Jung, Andrea Bianchi. **Design of Virtual Reality Application for Interaction Prototyping Remote Education**. In Proceedings of HCI Korea. The HCI Society of Korea, 2022. (HCIK '22) **Best Paper Award** ([video](#)) ([PDF](#))



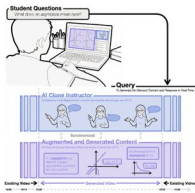
[C1] Neung Ryu, **Hye-Young Jo**, Michel Pahud, Mike Sinclair, Andrea Bianchi. **GamesBond: Bimanual Haptic Illusion of Physically Connected Objects for Immersive VR Using Grip Deformation**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2021. (CHI '21, acceptance rate: 26.3%) **Honorable Mention Award (top 5%)** ([video](#)) ([PDF](#))

Posters



[P1] **Hye-Young Jo**, Chan Hu Wie, Yejin Jang, Dong-Uk Kim, Yurim Son, Yoonji Kim. **TrainerTap: Weightlifting Support System Prototype Simulating Personal Trainer's Tactile and Auditory Guidance**. In Adjunct Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2023. (UIST '23 Adjunct, acceptance rate: 21.0%) ([video](#)) ([PDF](#))

Preprints



[U1] Hye-Young Jo, Ada Zhao, Xiaoan Liu, Ryo Suzuki. **Generative Lecture: Making Lecture Videos Interactive with LLMs and AI Clone Instructors**. arXiv preprint. 2025. ([video](#)) ([PDF](#))

Fellowship and Awards

Summary: I have received \$176,000 in fellowships, awards, and travel grants.

- Fall 2025 - Spring 2027* **Google Ph.D. Fellowship**
Awarded in the Human-Computer Interaction category; full tuition and stipend (\$85,000/year).
- 2026 **Conference Support Fund**
Awarded by the Department of Computer Science, University of Colorado Boulder (\$1,500).
- 2025 **Ralph J. Slutz Student Excellence Award**
Awarded by the Department of Computer Science, University of Colorado Boulder (\$1,000).
- 2024 **Conference Travel Grant**
Awarded by the Graduate and Professional Student Government, University of Colorado Boulder (\$500).
- 2024 **Gary Marsden Travel Award**
Awarded by ACM SIGCHI to attend CHI 2024 (\$3,000).
- 2023 **Best Master's Thesis Award**
Awarded to one of two master's theses in the Department of Industrial Design, KAIST.
- 2022 **Best Paper Award**
Awarded at HCI Korea 2022 [C2].
- 2022 **Top Research Award**
Awarded at the joint research seminar of Seoul National University, KAIST, Sogang University, and Korea University [X5].
- 2021 **Excellence Award**
Awarded at the Korea Metaverse Developer Contest 2021 [X5].
- 2021 **iF Design Award (Winner, UI/UX) and IDEA (Bronze, Digital Interaction)**
Contributed the information architecture to the control UI/UX of the Mobile Clinic Module, a negative-pressure isolation ward (team award, a multi-lab project of the Department of Industrial Design, KAIST).
- 2021 **Honorable Mention Award**
Awarded to the top 5% of submissions at CHI 2021 [C1].
- 2016 **Top Award**
Graduated first in class from the Maya 3D computer graphics course, Green Computer Academy.
- 2016 **Alumni Award**
Graduated second in class across all departments of the College of Fine Arts, Seoul National University.

Invited Talks and Lectures

Summary: I gave seven talks and guest lectures on AI-assisted video creation, prototyping, and my transition from art and film VFX to immersive media at companies and universities in the U.S. and Korea.

- August 2026* **Autodesk Research, Summer Internship Showcase**
Understanding Film Directing for AI-Assisted Video Creation
- October 2025* **Fujitsu Research of America, Lunch Seminar**
Grounding Generative Video in Real-World Geographies for Spatial Consistency
- May 2025* **KAIST, Make Lab Seminar**
Reimagining How We Create and Consume Content
- May 2025* **Seoul National University, Human-Centered Computing Systems Lab Seminar**
Reimagining How We Create and Consume Content
- February 2025* **University of Colorado Boulder, Guest Lecture in How to Hack Almost Anything (graduate course)**
Utilizing AI Services and APIs for Coding and Virtual Agent Creation
- October 2022* **KAIST, Guest Lecture in Interaction Prototyping (undergraduate course)**
DIY Arduino paper cases without 3D printing, using Adobe Illustrator, Autodesk Fusion 360, and Blender
- November 2019* **Kookmin University, Department of Entertainment Design Career Seminar**
Transitioning from Art and Film VFX to VR Content Creation

Teaching and Mentoring

Summary: I have taught introductory programming and physical prototyping as a teaching assistant, and mentored four undergraduate researchers building wearable haptic systems for fitness.

- Teaching Assistant**
- Fall 2025* Starting Computing, introductory programming in C++, University of Colorado Boulder, Instructor: Rhonda Hoenigman
- Fall 2022* Interaction Prototyping, physical prototyping with Arduino, KAIST
- Spring 2024* **Undergraduate Research Assistant**
Chan Hu Wie, Yejin Jang, Dong-Uk Kim, Yurim Son (Chung-Ang University)
Guided them through designing, building, and evaluating wearable haptic systems that recreate a personal trainer's touch, leading to a UIST poster [P1] and a CHI paper [C6].

Service

- 2025 - Present* **Reviewer**
CHI 2025-2026, DIS 2026

- Outreach and Leadership**
- 2025* University of Colorado Boulder Prospect Affiliate Match Program, helping prospective students prepare their PhD applications
- 2012* Seoul National University Mentoring Program, helping international students adjust to campus life

Media Coverage

October 2025

University of Colorado Boulder, Graduate School News
[“CU Boulder graduate student named a Google PhD fellow”](#)

May 2021

Microsoft Research Blog
[“Microsoft Research collaborates with KAIST in Korea to explore bimanual interactions with haptic feedback in virtual reality”](#)

Film VFX ([showreel](#))

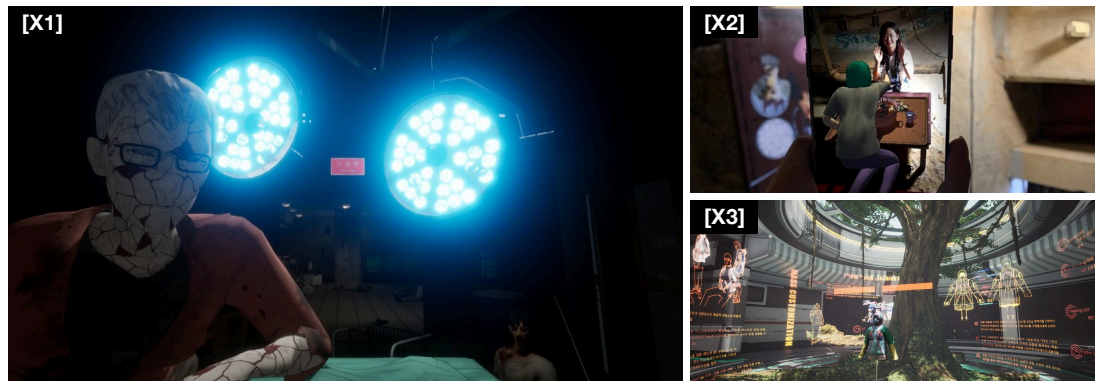
Summary: I worked as a visual effects (VFX) compositor on six feature films, three of which each drew over five million viewers in South Korea.



- 2018 [F6] <Along with the Gods: The Last 49 Days> ([VFX showreel](#)), Director: Yong-Hwa Kim. *Best Visual Effects, 2018 Grand Bell Awards. Over 12 million viewers in South Korea as of 2024.*
- 2017 [F5] <1987: When the Day Comes> ([VFX showreel](#)), Director: Jun-Hwan Jang. *Best Film, 2018 Blue Dragon Film Awards. Over 5 million viewers in South Korea as of 2024.*
- 2017 [F4] <Along with the Gods: The Two Worlds> ([VFX showreel](#)), Director: Yong-Hwa Kim. *Four awards including Best Technical Achievement, 2018 Blue Dragon Film Awards. Over 11 million viewers in South Korea as of 2024.*
- 2017 [F3] <Kung Fu Yoga> ([VFX showreel](#)), Director: Stanley Tong
- 2017 [F2] <Real> ([trailer](#)), Director: Sa-Rang Lee
- 2017 [F1] <Fabricated City> ([trailer](#)), Director: Kwang-Hyun Park

Immersive Content [\(showreel\)](#)

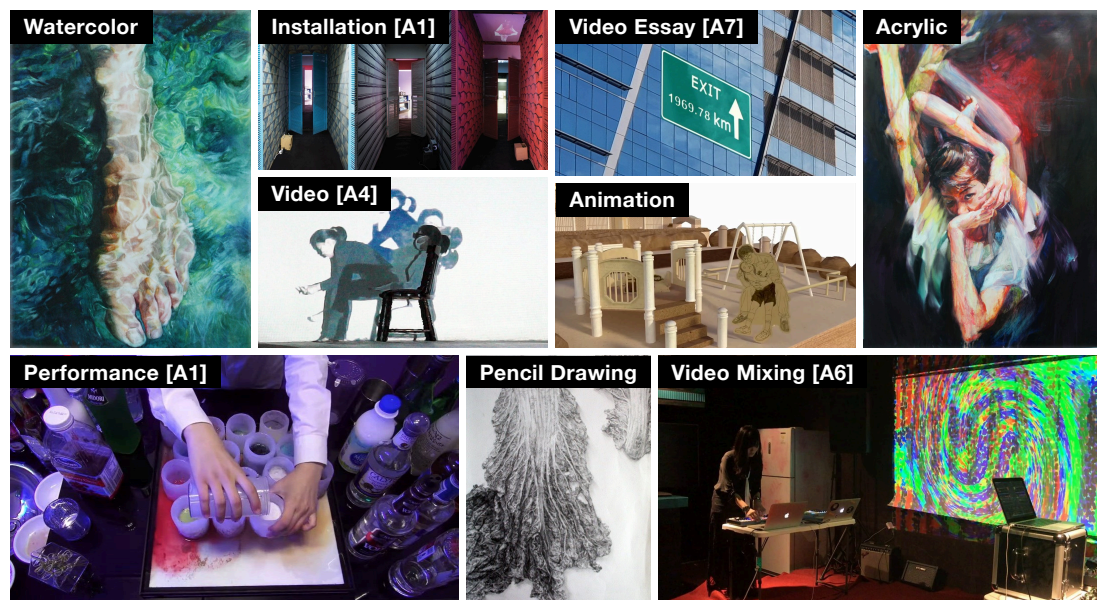
Summary: I created immersive experiences across VR and AR, including a VR horror film and AR exhibition, an AR app featuring miniature K-pop idols, a commissioned VR showroom for a tech-fashion exhibition, and a VR boxing game. I worked across concept design, 3D modeling, lighting, look development, coding, and camera operation.



- 2021 [X5] VR Boxing Game <Meta-Boxing> ([video](#)), Supervisor: Woontack Woo
- 2019 [X4] AR Mobile App <LGU+ 5G AR> ([video](#), [article](#)), Supervisor: Sun-Gu Kim
- 2019 [X3] VR Exhibition <Fashion For Help> ([video](#)), Supervisor: Young-Mo Son
- 2019 [X2] AR Exhibition <The Tide>, Supervisor: Sang-Hyoun Lee
- 2019 [X1] VR Toon Film <The Tide> ([Steam](#)), Director: Tae-Kyung Yoo. *Official selection, New Frontier, 2019 Sundance Film Festival.*

Art Exhibitions and Live Performances [\(showreel\)](#)

Summary: My art explores the human body, emotion, and relationships between people through identity and belonging, the precarity of young adults in Korea, and fear and isolation. Across painting, video, installation, and performance, I am drawn to real-time and participatory work that audiences help complete, from a one-on-one cocktail bar to live video mixing.



- 2018 [A7] Group Exhibition <Lapses>, Platform-L Contemporary Art Center, Curator: Eobchae (funded by Hyundai's ZER01NE project)
- 2016 [A6] Live Video Jockey Performance <We Play>, Sangsangmadang, Supervisor: Hoon-Gyu Park (Parkpunk)
- 2016 [A5] Invited Exhibition <Don't fake it, believe it>, Myhttake Museum
- 2015 [A4] Group Exhibition <Dirt Luv for Graduation> (videos: [Package for Me](#), [Human-furniture](#), [Walking on the Spot](#)), Seoul National University
- 2015 [A3] Group Exhibition <8-bit>, Seoul National University
- 2015 [A2] Group Exhibition <Crawling>, Seoul National University
- 2014 [A1] Group Exhibition <The Great Exodus> (videos: [Closet Inside the Closet](#), [1:1 Bar](#)), Seoul National University. *One of three finalists for the Brighton Prize.*